The University of Texas at Austin
Solutions to In-Term Exam 3

M378K Introduction to Mathematical Statistics

April 25, 2026

Instructions: This is a closed book and closed notes exam. The maximal score on the exam is 100 points.
Time: 50 minutes

Formulas.

If Y has the binomial distribution with parameters n and p, then $p_Y(k) = \mathbb{P}[Y = k] = \binom{n}{k} p^k (1-p)^{n-k}$, for $k = 0, ..., n$, $\mathbb{E}[Y] = np$, Var $[Y] = np(1-p)$. The binomial coefficients are defined as follows for integers $0 \le k \le n$, $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. The moment generating function of Y is given by $m_Y(t) = \left(pe^t + q\right)^n$.

If Y has a geometric distribution with parameter p, then $p_Y(k) = p(1-p)^k$ for $k = 0, 1, ...$, $\mathbb{E}[Y] = \frac{1-p}{p}$, Var $[Y] = \frac{1-p}{p^2}$. Its mgf is $m_Y(t) = \frac{p}{1-qe^t}$ for t such that $qe^t < 1$.

If Y has a Poisson distribution with parameter λ, then $p_Y(k) = e^{-\lambda}\frac{\lambda^k}{k!}$ for $k = 0, 1, ...$, $\mathbb{E}[Y] =$ Var $[Y] = \lambda$. Its mgf is $m_Y(t) = e^{\lambda(e^t - 1)}$.

If Y has a uniform distribution on $[l, r]$, its density is

$$f_Y(y) = \frac{1}{r-l} 1_{(l,r)}(y),$$

its mean is $\frac{l+r}{2}$, and its variance is $\frac{(r-l)^2}{12}$. Let $U \sim U(0,1)$. The mgf of U is $m_U(t) = \frac{1}{t}(e^t - 1)$.

If Y has the standard normal distribution, then its mean is zero, its variance is one, and its density equals

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad x \in \mathbb{R}.$$

Its mgf is $m_Y(t) = e^{\frac{t^2}{2}}$.

If Y has the exponential distribution with parameter τ, then its cumulative distribution function is $F_Y(y) = 1 - e^{-\frac{y}{\tau}}$ for $y \ge 0$, its probability density function is $f_Y(y) = \frac{1}{\tau} e^{-y/\tau}$ for $y \ge 0$. Also, $\mathbb{E}[Y] = SD[Y] = \tau$. Its mgf is $m_Y(t) = \frac{1}{1-\tau t}$.

The mgf of $Y \sim \Gamma(k, \tau)$ is

$$m_Y(t) = \frac{1}{(1-\tau t)^k} \text{ for } \quad t < 1/\tau.$$

Its expectation is $k\tau$ and its variance is $k\tau^2$. The χ^2 —distribution with n degrees of freedom is the special case $\Gamma(\frac{n}{2}, 2)$

DEFINITIONS

Problem 3.1. (10 points) Write down the definition of the **bias** of an estimator $\hat{\theta}$ of a parameter θ.

Solution.

$$\text{bias}\left(\hat{\theta}\right) = \mathbb{E}\left[\hat{\theta}\right] - \theta$$

Problem 3.2. (10 points) Write down the definition of the **mean squared error** of an estimator $\hat{\theta}$ of a parameter θ.

Solution.

$$\text{MSE}\left(\hat{\theta}\right) = \mathbb{E}\left[\left(\hat{\theta} - \theta\right)^2\right]$$

TRUE/FALSE QUESTIONS

Problem 3.3. (5 points) Let Z be a standard normal random variable and let Q^2 have the χ^2 —distribution with ν degrees of freedom. Assume that Z and Q^2 are independent. Set

$$T = \frac{Z}{\sqrt{\frac{Q^2}{\nu}}}.$$

Then, T has a t —distribution with ν degrees of freedom. *True or false?*

Solution. TRUE
From the definition of the t —distribution, we know that

$$T = \frac{Z}{\sqrt{\frac{Q^2}{\nu}}}$$

has the t — distribution with ν degrees of freedom.

Problem 3.4. (5 points) The p-value of the test can be controlled by the statistician by choosing different level of significance α. *True or false? Why?*

Solution. FALSE
By definition, the p —value is the probability of observing a value of the test statistic or something "more extreme". It has nothing to do with the significance level.

Problem 3.5. (5 points) If a null hypothesis fails to be rejected at $\alpha = 0.05$, it will also fail to be rejected at $\tilde{\alpha} = 0.01$. *True or false? Why?*

Solution. TRUE
By the hypothesis testing procedure, we fail to reject the null hypothesis if the p —value exceeds the significance level. In this case, since the p —value exceeds 0.05, it must also exceed 0.01.

Free-response problems.

Please, explain carefully all your statements and assumptions. Numerical results or single-word answers without an explanation (even if they're correct) are worth 0 points.

Problem 3.6. (20 points) Let $Y_1, Y_2, \ldots, Y_n$ be a random sample from a population with a uniform distribution on $(\theta, \theta + 1)$. Then,

$$\hat{\theta}_n = \bar{Y}_n - \frac{1}{2}$$

is a **consistent** estimator for θ. *True or false? Why?*

Solution. We can use the same theorem we used in class, i.e., we can demonstrate that

- $\hat{\theta}_n$ is *unbiased*; and
- $\text{Var}\left[\hat{\theta}_n\right] \to 0$, as $n \to \infty$.

To show that $\hat{\theta}_n$ is unbiased, we must prove that

$$\mathbb{E}\left[\hat{\theta}_n\right] = \theta.$$

From the given definition of $\hat{\theta}_n$, we have

$$\mathbb{E}\left[\hat{\theta}_n\right] = \mathbb{E}\left[\bar{Y}_n - \frac{1}{2}\right] = \mathbb{E}[Y_1] - \frac{1}{2} = \theta + \frac{1}{2} - \frac{1}{2} = \theta.$$

Hence, $\hat{\theta}_n$ is, indeed, unbiased.

As for the other claim, we have that

$$\text{Var}\left[\hat{\theta}_n\right] = \text{Var}\left[\bar{Y}_n - \frac{1}{2}\right] = \text{Var}\left[\bar{Y}_n\right].$$

Since $Y_1, \ldots, Y_n$ is a random sample, the additive formula for the variance applies. So, we get

$$\text{Var}\left[\hat{\theta}_n\right] = \frac{\text{Var}\left[Y_1\right]}{n} = \frac{1}{12n}.$$

The above converges to 0 as $n \to \infty$ which concludes our proof.

Problem 3.7. (20 points) Let $Y_1, \ldots, Y_n$ be a random sample from a distribution with following density function:

$$f^{\theta}(y) = (\theta + 1) y^{\theta} 1_{()[0,1]}(y)$$

for an unknown parameter $\theta > -1$. What is the maximum likelihood estimator for θ? Make sure that you prove your claim!

Solution. See https://mcudina.github.io/page/M378K/notes/examples_MLE_MM_notes.pdf

MULTIPLE CHOICE QUESTIONS

Problem 3.8. (10 points) Let $(Y_1, \ldots, Y_n)$ be a random sample from the exponential distribution $E(\tau)$ with the unknown parameter $\tau > 0$. Which of the following is a pivotal quantity? Circle **all** that apply!

a. $\frac{1}{\tau}\bar{Y}$
b. $\frac{1}{\tau}Y_1$
c. $\tau^{-n}\prod Y_i$
d. τ/Y_n
e. <u>None</u> of the above

Solution. The correct answer is **(a), (b), (c), (d)**.
All of the above are functions of the random variables $W_1 = \frac{1}{\tau}Y_1, ..., W_n = \frac{1}{\tau}Y_n$, each of which has the $E(1)$-distribution. The distributions of $W_1, ..., W_n$ clearly do not depend on τ, and, therefore, neither do the distributions of their functions $\bar{W}$, W_1, $\prod W_i$ and $1/W_n$.

Problem 3.9. (5 points) In a random sample of 100 voters 64 prefer candidate A and the rest prefer candidate B. The (approximate) $(1-\alpha)$-confidence interval for the parameter p (the population proportion of A voters) is of the form

$$\left[0.64 - z_{\alpha/2} \times c, 0.64 + z_{\alpha/2} \times c\right],$$

where $z_{\alpha/2} = \texttt{qnorm}\ (1-\alpha/2, 0, 1)$.

The value of c is:

a. 0.016
b. 0.024
c. 0.036
d. 0.048
e. ≥ 0.05

Solution. The correct answer is **(d)** since $c = \sqrt{\hat{p}(1-\hat{p})/n} = \sqrt{0.64 \times 0.36/100} = 0.048$.

Problem 3.10. (5 points) A sample of size $n = 2$ from normal distribution with unknown μ and σ is collected and the data are

$$y_1 = 1 \quad \text{and} \quad y_2 = 5.$$

The right end-point of a symmetric 95% confidence interval for σ^2 is

a. 8/`qchisq(0.975,2)`
b. 16/`qchisq(0.025,1)`
c. 8/`qchisq(0.025,1)`
d. 16/`qchisq(0.975,2)`
e. None of the above

Solution. The correct answer is **(c)**.
The confidence interval is based on the pivotal quantity $(n-1)S^2/\sigma^2$ whose distribution is $\chi^2(n-1)$. In this case, $n = 2$, $\bar{Y} = 3$ so that $(n-1)s^2 = (y_1 - \bar{y})^2 + (y_2 - \bar{y})^2 = 2^2 + 2^2 = 8$, which produces the interval

$$[8/\texttt{qchisq}(0.975, 1), 8/\texttt{qchisq}(0.025, 1)].$$

Problem 3.11. (5 points) Let $Y_1, ..., Y_5$ be a random sample from the normal distribution $N(\mu, \sigma)$, with an <u>unknown</u> mean μ and an <u>unknown</u> standard deviation σ. The collected data turn out to have, in our usual notation,

$$\sum_{i=1}^{5} y_i = 15 \quad \text{and} \quad \sum_{i=1}^{5} (y_i - 3)^2 = 9.$$

The <u>right</u> end-point of the symmetric 90%-confidence interval $[\hat{\mu}_L, \hat{\mu}_R]$ for μ is

a. $3 + \frac{3}{2\sqrt{5}}$ `qt` $(0.95, 4)$
b. $3 - \frac{\sqrt{5}}{\sqrt{8}}$ `qt` $(0.1, 4)$.
c. $3 - \frac{3}{2\sqrt{5}}$ `qt` $(0.95, 4)$.
d. $3 - \frac{1}{\sqrt{2}}$ `qt` $(0.1, 4)$.
e. None of the above.

Solution. The correct answer is **(a)**.
The confidence interval is based on the pivotal quantity $\frac{\bar{Y}-\mu}{\sqrt{S^2/n}}$ which has the t −distribution with 4 degrees of freedom. Therefore, for $b =$ `qt` $(0.95, 4)$ we have

$$\mathbb{P}\left[-b \leq \frac{(\bar{Y} - \mu)}{\sqrt{S^2/n}} \leq b\right] = 0.90.$$

We "solve" for μ to obtain

$$\mathbb{P}\left[\bar{Y} - b\sqrt{S^2/n} \leq \mu \leq \bar{Y} + b\sqrt{S^2/n}\right] = 0.90.$$

For our data set $\bar{y} = 3$ and $S^2 = 9/4$, so $\hat{\mu}_R = 3 + \frac{3}{2\sqrt{5}}$ `qt` $(0.95, 4)$.